

Lecture 9 - Part II

Costs of production

ECON 101

Fall 2020



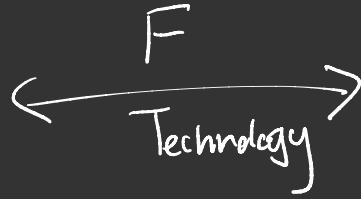
Duality between production & costs

Moral: we take as primitive either the production function or the cost function

Production function

$$q = F(K, L)$$

How much q given K, L



(Total) Cost Function

$$\text{Cost} = C(q, \text{input prices})$$

How costly is to produce q given some inputs

K = units of capital

L = hours of labor

$$\text{Cost} = wL + rK$$

↑
wage/hour

↑
cost of capital
(rental rate)

The goal of the firm is to choose inputs optimally so as to produce q output

We will distinguish between:

- Short-Run Costs : at least one input is fixed
 - Long-Run Costs : all inputs are variable.
-

We start with the short-run.

Example:

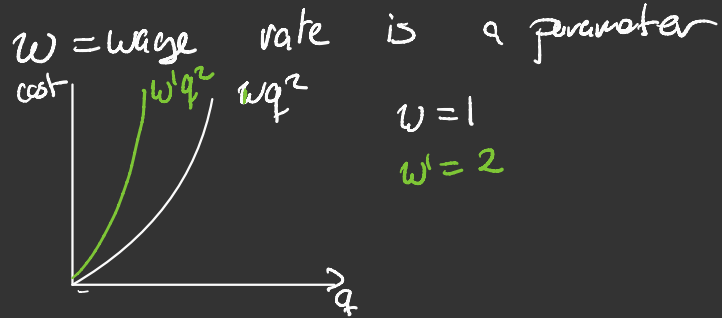
One-input production function $q = \sqrt{L}$

To produce q how many units of labor I need?

$q = \sqrt{L} \Rightarrow q^2 = L$ this is how much I need.

$$\Rightarrow \text{Cost} = wL = wq^2$$

$$C(q, w) = wq^2 \Rightarrow$$



Suppose we have a two-input production function $q = F(K, L)$

Example: $q = A K^{1/2} L^{1/2}$

Short-Run say $\bar{K} = 100$ units, its per unit cost is $r = 1$

If I want to produce q I will need amount of L equal to

$$q = A (100)^{1/2} L^{1/2} = A 10 L^{1/2} \Rightarrow \left(\frac{q}{A 10} \right)^2 = L$$

$$L = \frac{q^2}{A^2 100}$$

Costs are:

$$C = wL + r \bar{K} = w \left(\frac{q^2}{A^2 100} \right) + (1) 100$$

Short-Run fixed

$$C(q, w) = \frac{w q^2}{100 A^2} + 100 ;$$

Note A is a technology parameter

In general the Total cost is:

$$TC = VC + FC$$

↑
Variable cost

Fixed cost.

Example: $TC = \frac{wq^2}{100A^2} + 100$

$$VC(q, w) = \frac{wq^2}{100A^2}, \quad \underline{FC = 100}$$

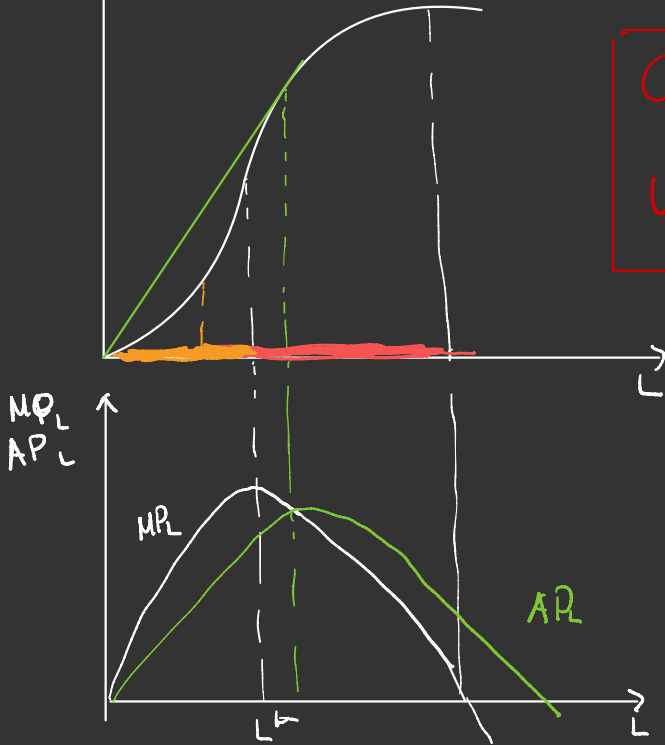
variable cost, depends on q

Fixed cost, no dependence on q

Example:

Recall: Cubic production functions:

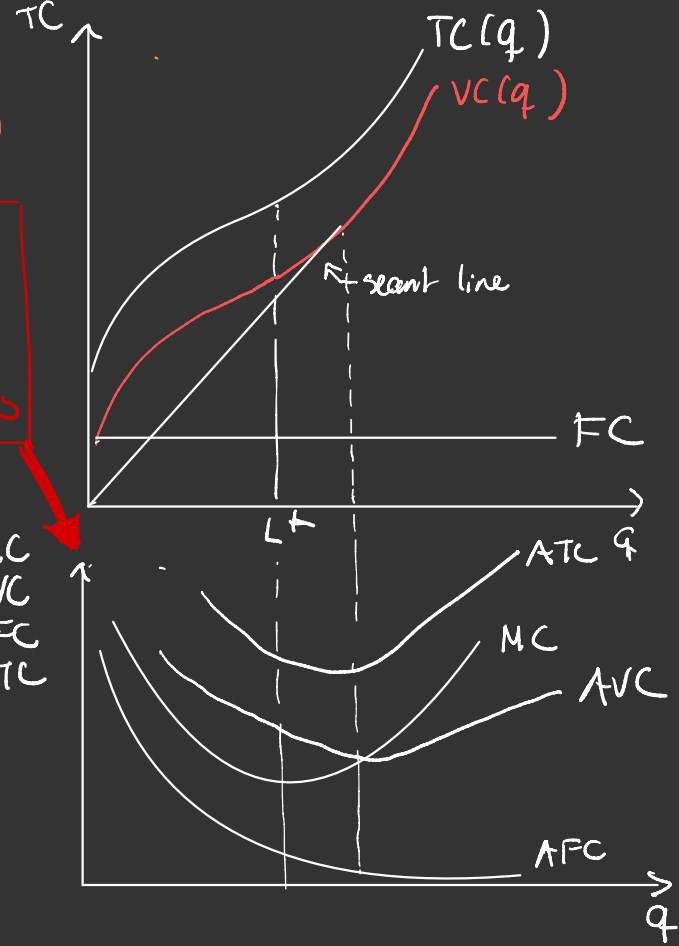
$f(L)$



MP_L is increasing
 MP_L is decreasing

Cubic costs
generate
U-shaped
Cost Functions

gives rise → Cubic Costs



Example: $q = A K^\alpha L^{1-\alpha}$

$\alpha = \frac{1}{2}$; $w = 1$; $r = 1$; $K = 100$; $A = 1$

$$\left. \begin{array}{l} \alpha = \frac{1}{2} ; w = 1 \\ \text{Short Run: } K = 100 ; r = 1 \\ A = 1 \end{array} \right\} q = 10 \sqrt{L}$$

$\Rightarrow \frac{q^2}{100} = L \Rightarrow \text{Total Cost is}$

$$TC(q) = \underbrace{(1) \left(\frac{q^2}{100} \right)}_{wL} + \underbrace{(1)(100)}_{rK}$$

$$TC(q) = \frac{q^2}{100} + 100$$

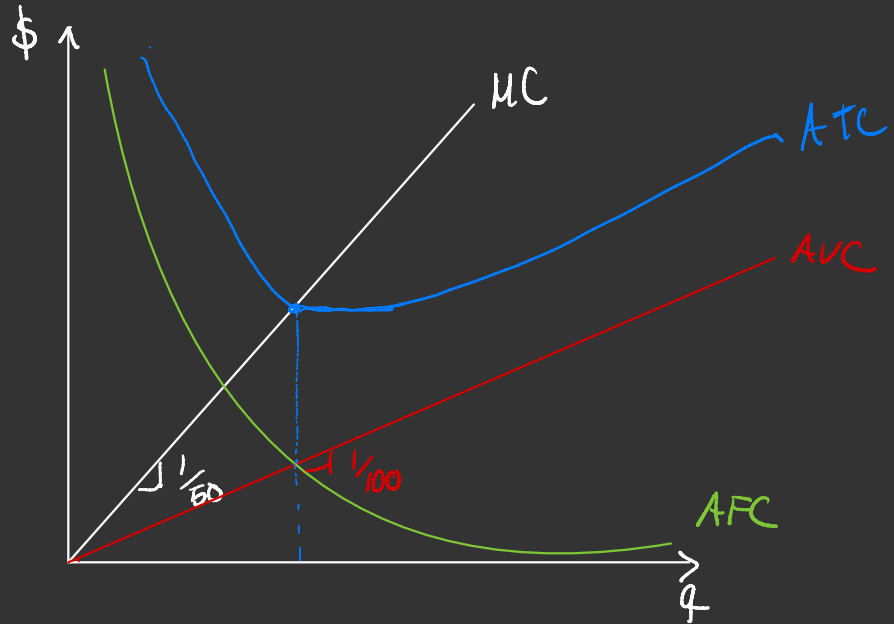
$$VC(q) = \frac{q^2}{100}, FC = 100$$

giving to you: $MC(q) = \frac{q}{50}$

$$AVC(q) = \frac{VC(q)}{q} = \frac{q}{100}$$

$$AFC(q) = \frac{FC}{q} = \frac{100}{q}$$

$$ATC(q) = \frac{q}{100} + \frac{100}{q}$$



Costs in the Long-Run

- All inputs are variable $\Rightarrow$ goal of the firm is to choose optimally its inputs of production so as to minimize cost, given a target level of output

$$C(w, r, q) = wL^* + rK^* = \underset{L, K}{\text{minimize}} \quad wL + rK$$

Long run cost function.

s.t.

$$q = F(L, K)$$

$\downarrow$ optimal solutions

$$L^*(w, r, q)$$

$$K^*(w, r, q)$$

optimal level of inputs that minimize cost at prices (w, r) and target q

Dual relationship between production function & cost function.

Thus Cost function is sometimes used as the primitive

w = wage rate

r = rental rate

F is the prod. function

q = target level of output

(w, r, q) are parameters

This problem is actually very similar to the (alternative) problem of the consumer (expenditure minimization)

Objective:

in the minimization problem

$C = wL + rK$, so let's consider a fixed cost of $\bar{C}$

The set of all input combinations (L, K) that achieve cost $\bar{C}$ is given by the expression

$$\bar{C} = wL + rK$$

↓ eq. form

$$K = \frac{\bar{C}}{r} - \frac{w}{r}L$$

$\frac{w}{r}$ = input price ratio

The iso-cost line is what moves in this problem!

Iso-cost line

Iso: greek for "the same"

What about the constraint? $q = F(L, K)$

$q_0 = F(L, K)$ defines implicitly an isoquant

↳ all the (L, K) -input combinations that satisfy this equation give the exact same level of output

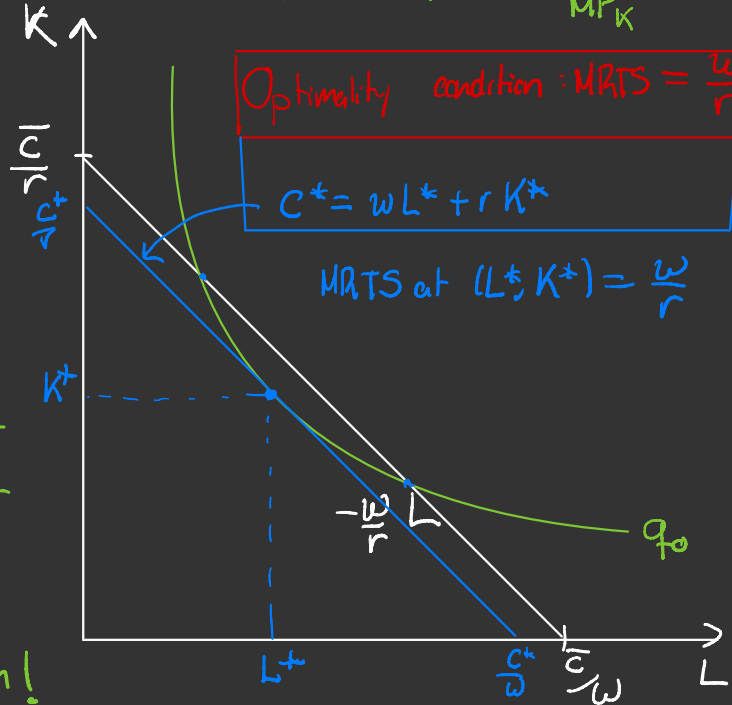
The target isoquant is fixed in this problem!

slope of isoquant = $-\frac{MP_L}{MP_K} = -MRTS$

Optimality condition: $MRTS = \frac{w}{r}$

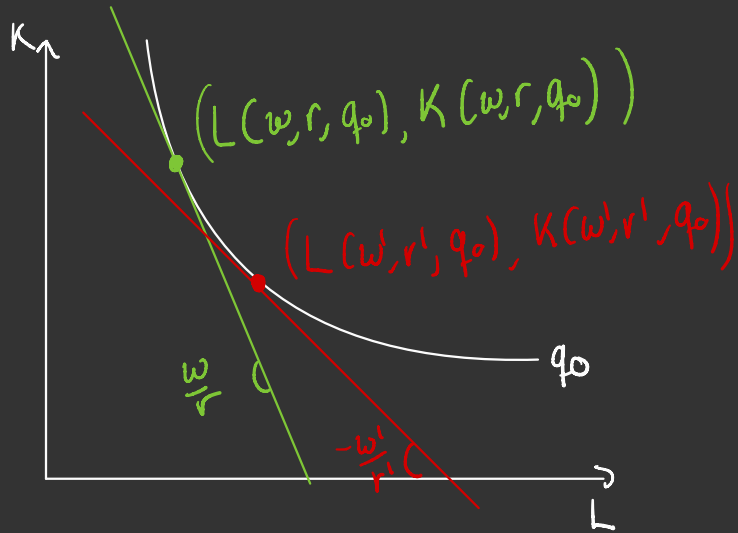
$$C^* = wL^* + rK^*$$

$MRTS \text{ at } (L^*, K^*) = \frac{w}{r}$



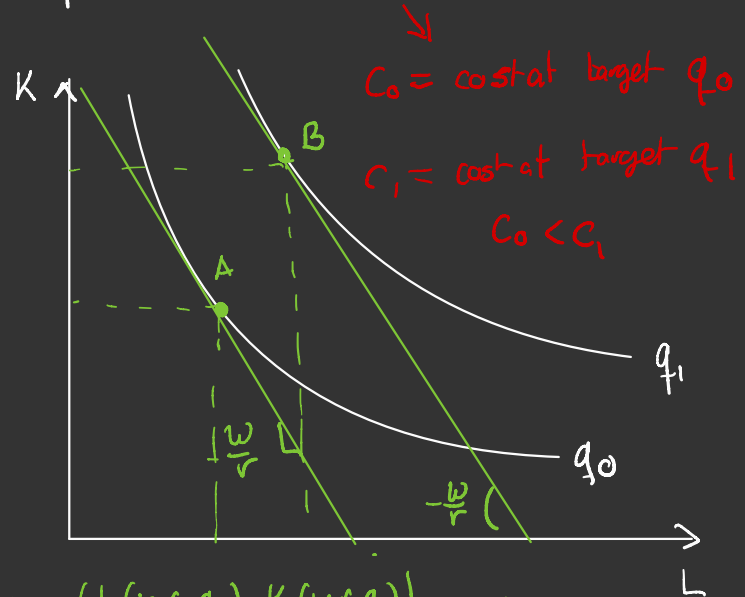
What if I change prices w, r
but not quantity?

$$\frac{w}{r} \rightarrow \frac{w'}{r'}$$



What happens if w, r remain fixed but
the target output changes:

$$q_0 \rightarrow q_1, \quad q_0 < q_1$$



$$A = (L(w, r, q_0), K(w, r, q_0)) \rightarrow wL(w, r, q_0) + rK(w, r, q_0) = C_0$$

$$B = (L(w, r, q_1), K(w, r, q_1)) \rightarrow wL(w, r, q_1) + rK(w, r, q_1) = C_1$$

Example :

$$\begin{aligned} &\text{minimize}_{L, K} \quad wL + rK \\ &q = L^{1/2} K^{1/2} \end{aligned}$$

$$MRTS = \frac{K}{L}$$

Optimality condition: Expansion path

$$\frac{K}{L} = \frac{w}{r} \Rightarrow \boxed{K = \frac{w}{r} L}$$

Plug into the constraint-

$$q = L^{1/2} \left(\frac{w}{r} L \right)^{1/2}$$

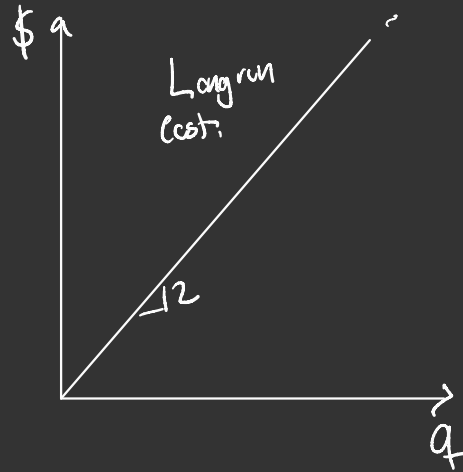
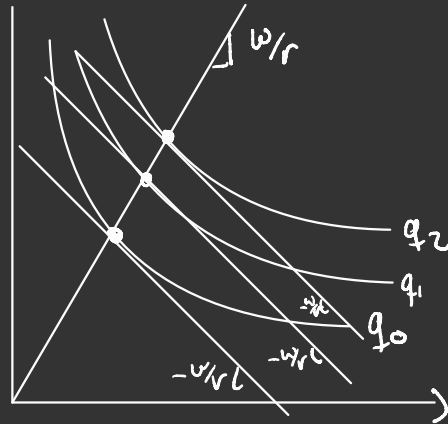
$$\Rightarrow q = L \sqrt{\frac{w}{r}}$$

$$\begin{aligned} L^+(w, r, q) &= q \sqrt{\frac{r}{w}} \\ K^+(w, r, q) &= q \sqrt{\frac{w}{r}} \end{aligned}$$

Optimal Cost (Long-Run) :

$$\begin{aligned} C(w, r, q) &= wq \sqrt{\frac{r}{w}} + rq \sqrt{\frac{w}{r}} \\ &= q \left(w \frac{\sqrt{r}}{\sqrt{w}} + r \frac{\sqrt{w}}{\sqrt{r}} \right) \\ &= q \left(\sqrt{w} \sqrt{r} + \sqrt{r} \sqrt{w} \right) \\ &= 2q \sqrt{wr} \end{aligned}$$

suppose $w=r=1$
 $\Rightarrow C(q) = 2q$



Technological change:

$$q = A L^{1/2} K^{1/2}$$

$$q_0 = A L^{1/2} K^{1/2}$$

$$\begin{array}{ll} \text{minimize} & wL + rK \\ L, K & \text{s.t. } q_0 = A L^{1/2} K^{1/2} \end{array}$$

$$\text{Optimality: } \frac{K}{L} = \frac{w}{r} \Rightarrow K = \frac{w}{r} L$$

$$q_0 = A L^{1/2} \left(\frac{w}{r} L \right)^{1/2}$$

$$q_0 = A L \sqrt{\frac{w}{r}}$$

$$L^* = \frac{q_0 \sqrt{\frac{r}{w}}}{A} \quad K^* = \frac{q_0 \sqrt{\frac{w}{r}}}{A}$$

$$\text{Long-run: } C(w, r, q_0) = \frac{2q_0}{A} \sqrt{wr}$$

$$\text{If } A \uparrow \rightarrow C \downarrow$$